



Lecture 19

In the last class we found the peak time $t_p = \frac{\pi}{\omega_d}$.

Now we would like to find the other quantity of interest.

Overshoot occurs at the peak time t_p . By using the relation found for t_p , we can find an expression for M_p .

$$y(t) = 1 - e^{-\zeta \omega_n t} \left(\cos \omega_d t + \frac{\zeta \omega_n}{\omega_d} \sin \omega_d t \right)$$

$$t_p = \frac{\pi}{\omega_d}$$

$$y(t_p) = 1 + M_p = 1 - e^{-\zeta \omega_n t_p} (\cos \omega_d t_p$$

$$+ \frac{\xi \omega_n}{\omega_d} \sin \omega_d t_p)$$

$$= 1 - e^{-\xi \omega_n t_p} \left(\cos \cancel{\omega_d} \cdot \cancel{\frac{\pi}{\omega_d}} + \frac{\xi \omega_n}{\omega_d} \sin \cancel{\omega_d} \cdot \cancel{\frac{\pi}{\omega_d}} \right)$$

$$= 1 + e^{-\cancel{\xi \omega_n} \cdot \frac{\pi}{\cancel{\omega_n \sqrt{1-\xi^2}}}}$$

$$1 + M_p = 1 + e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}}$$

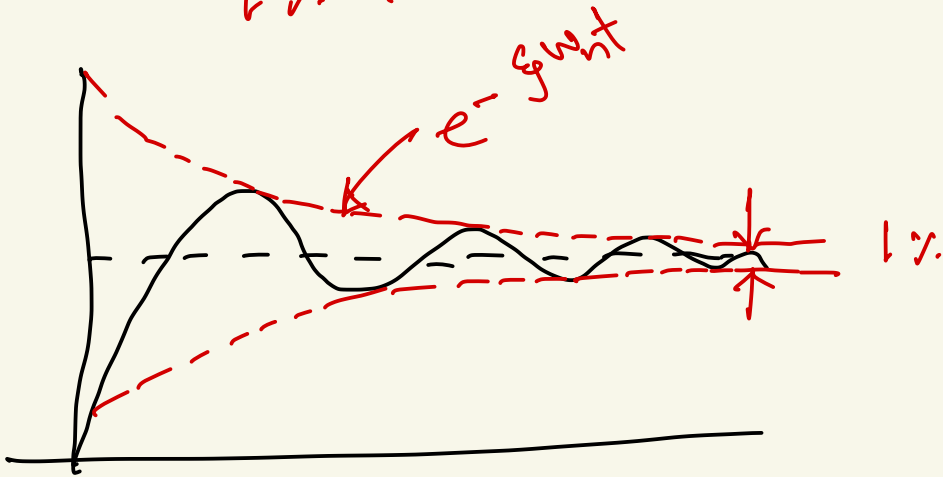
$$\Rightarrow M_p = e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}}, \quad 0 \leq \xi < 1$$

$$\Rightarrow \% M_p = e^{-\frac{\xi \pi}{\sqrt{1-\xi^2}}} \times 100 \%$$

For settling time t_s , the response has to be within 1% of the

final value.

$$1\% = 0.01$$



$$e^{-\zeta \omega_n t_s} = 0.01$$

$$\Rightarrow -\zeta \omega_n t_s = \ln 0.01$$

$$\Rightarrow -\zeta \omega_n t_s = -4.6$$

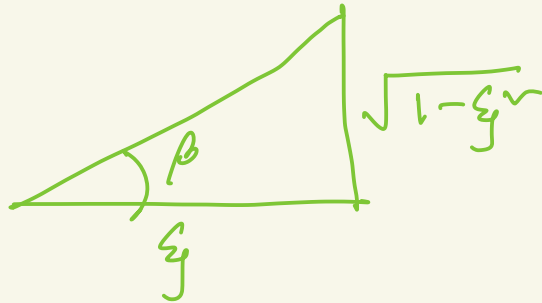
$$\Rightarrow t_s = \frac{4.6}{\zeta \omega_n} = \frac{4.6}{\sigma}$$

Now only parameter remaining is the rise time. In order to

compute that we need to slightly modify the step response.

$$y(t) = 1 - e^{-\zeta \omega_n t} \left(\cos \omega_d t + \frac{\zeta \omega_n}{\omega_d} \sin \omega_d t \right) \\ = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \left(\sqrt{1-\zeta^2} \cos \omega_d t + \zeta \sin \omega_d t \right)$$

$$\sin \beta = \sqrt{1-\zeta^2}, \quad \cos \beta = \zeta$$



$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

we can write compactly as follows

$$y(t) = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \beta)$$

For rise time, $y(t_r) = 1$

$$1 = 1 - \frac{e^{-\zeta \omega_n t}}{\sqrt{1-\zeta^2}} \sin(\omega_d t + \beta)$$

$$\Rightarrow \sin(\omega_d t_r + \beta) = 0 = \sin k\pi$$

$k=1, 2, \dots$

$$\Rightarrow \sin(\omega_d t_r + \beta) = \pi \quad k=1$$

$$\Rightarrow \omega_d t_r + \beta = \pi$$

$$\Rightarrow \omega_d t_r = \pi - \beta$$

$$t_p = \frac{\pi - \beta}{\omega_d}$$

So, we have found all the required relationship.

$$t_p = \frac{\pi}{\omega_d}$$

$$M_p = e^{-\pi \zeta / \sqrt{1 - \zeta^2}}$$

$$t_s = \frac{4.6}{\zeta \omega_n} = \frac{4.6}{\sigma}$$

$$t_p = \frac{\pi - \beta}{\omega_d}$$

